

K2-29b

R-2027

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_241211 · created 2026-08-08T00:50:58+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460655.6539 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.029
Transit depth (Rp/Rs)^2	1.57 +/- 0.71 [%]
Orbital Inclination (inc)	89.41 +/- 1.61
Airmass coefficient 1 (a1)	1.001 +/- 0.01
Airmass coefficient 2 (a2)	-0.005 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#3 - [279, 340]
Optimal Aperture	3.52
Optimal Annulus	11.36
Transit Duration (day)	0.0753 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.029 vs 0.1419 (deviation 0.58σ)
Duration fit vs published	0.0753 d vs 0.0925 d (deviation 2.32σ)
Mid-time O–C	-18.6 min
Depth SNR	4.3
Residual scatter	1.02 %

Light curve

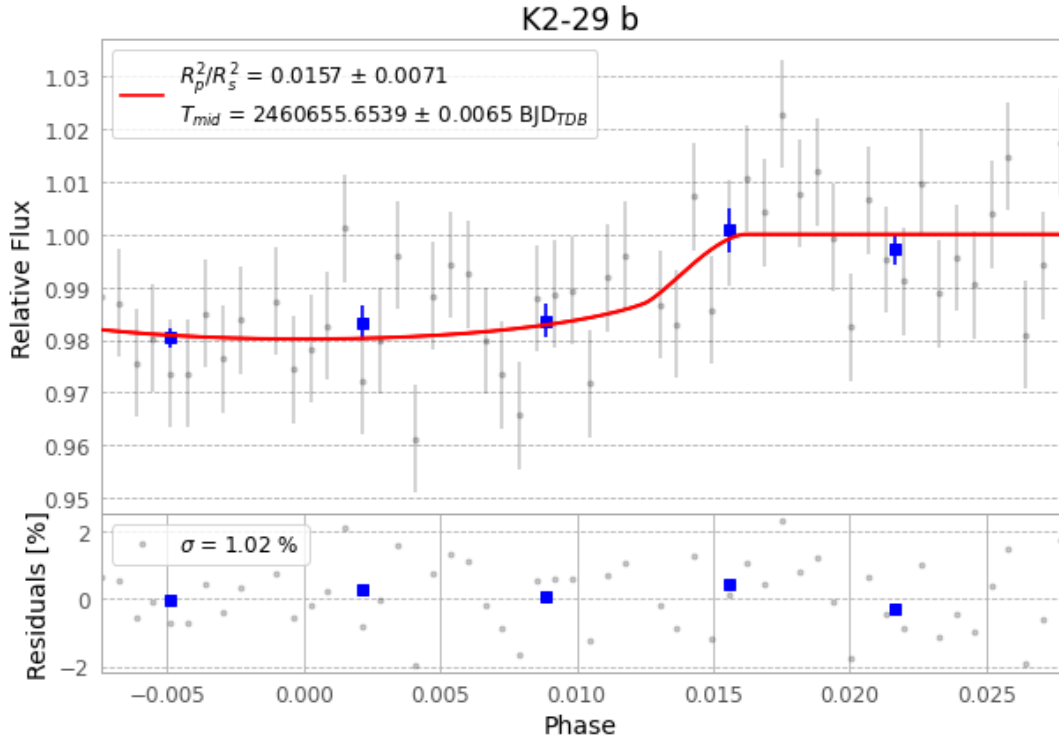


Figure 1 — FinalLightCurve_K2-29 b_2024-12-10.png (SHA-256 142d9092d07fbce0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aec2abbb204a4cdf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

56 FITS frames (37 MB), K2-29241211025715.FITS → K2-29241211054216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-241211.log (a812a014b251...), FinalLightCurve_K2-29 b_2024-12-10.png (142d9092d07f...), FinalLightCurve_K2-29 b_2024-12-10.csv (f3385e34f9e2...)

Compute resources

Wall time 102.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:50Z.